

Measures of Position

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- Percentiles
- Deciles and Quartiles
- The Box Plot

Measures of Position

In some cases we are interested in certain individual items in the data set, rather than in the set as a whole. We need a way of measuring how an item fits into the collection, how it compares to other items in the collection, or even how it compares to another item in another collection. There are several common ways of creating such measures and they are usually called **measures of position**.

The z-Score

If x is a data item in a sample with mean $\bar{x}$ and standard deviation s , then the **z-score** of x is given by

$$z = \frac{x - \bar{x}}{s}.$$

Example: Comparing with z-Scores

Two students, who take different history classes, had exams on the same day. Jen's score was 83 while Joy's score was 78. Which student did relatively better, given the class data shown below?

	Jen	Joy
Class mean	78	70
Class standard deviation	4	5

Example: Comparing with z-Scores

Solution

Calculate the z-scores:

$$\text{Jen: } z = \frac{83 - 78}{4} = 1.25 \qquad \text{Joy: } z = \frac{78 - 70}{5} = 1.6$$

Since Joy's z-score is higher, she was positioned relatively higher within her class than Jen was within her class.

Percentiles

When you take a standardized test taken by larger numbers of students, your raw score is usually converted to a **percentile** score, which is defined on the next slide.

Percentiles

If approximately n percent of the items in a distribution are less than the number x , then x is the **n th percentile** of the distribution, denoted P_n .

Example: Percentiles

The following are test scores (out of 100) for a particular math class.

44 56 58 62 64 64 70 72
72 72 74 74 75 78 78 79
80 82 82 84 86 87 88 90
92 95 96 96 98 100

Find the fortieth percentile.

Example: Percentiles

Solution

The 40th percentile can be taken as the item below which 40 percent of the items are ranked. Since 40 percent of 30 is $(.40)(30) = 12$, we take the thirteenth item, or 75, as the fortieth percentile.

Deciles and Quartiles

Deciles are the nine values (denoted $D_1, D_2, \dots, D_9$) along the scale that divide a data set into ten (approximately) equal parts, and **quartiles** are the three values (Q_1, Q_2, Q_3) that divide the data set into four (approximately) equal parts.

Example: Deciles

The following are test scores (out of 100) for a particular math class.

44 56 58 62 64 64 70 72
72 72 74 74 75 78 78 79
80 82 82 84 86 87 88 90
92 95 96 96 98 100

Find the sixth decile.

Example: Percentiles

Solution

The sixth decile is the 60th percentile. Since 60 percent of 30 is $(.60)(30) = 18$, we take the nineteenth item, or 82, as the sixth decile.

Finding Quartiles

For any set of data (ranked in order from least to greatest):

The **second quartile**, Q_2 , is just the median.

The **first quartile**, Q_1 , is the median of all items below Q_2 .

The **third quartile**, Q_3 , is the median of all items above Q_2 .

Example: Quartiles

The following are test scores (out of 100) for a particular math class.

44 56 58 62 64 64 70 72
72 72 74 74 75 78 78 79
80 82 82 84 86 87 88 90
92 95 96 96 98 100

Find the three quartiles.

Example: Percentiles

Solution

The two middle numbers are 78 and 79 so

$$Q_2 = (78 + 79)/2 = 78.5.$$

There are 15 numbers above and 15 numbers below Q_2 , the middle number for the lower group is

$$Q_1 = 72, \text{ and for the upper group is}$$

$$Q_3 = 88.$$

The Box Plot

A **box plot**, or **box-and-whisker plot**, involves the median (a measure of central tendency), the range (a measure of dispersion), and the first and third quartiles (measures of position), all incorporated into a simple visual display.

The Box Plot

For a given set of data, a **box plot** (or **box-and-whisker plot**) consists of a rectangular box positioned above a numerical scale, extending from Q_1 to Q_3 , with the value of Q_2 (the median) indicated within the box, and with “whiskers” (line segments) extending to the left and right from the box out to the minimum and maximum data items.

Example:

Construct a box plot for the weekly study times data shown below.

1	5	8			
2	0	7	8	9	9
3	2	6	6	7	
4	0	2	2	7	9
5	1	5	6		
6	6				

Example:

Solution

The minimum and maximum items are 15 and 66.

